ModelIt!

Student Activity Pack

Genetic Mutations: When DNA Makes a Typo

How UV Radiation and Cell Division Create Errors That Change Organisms

Grade 08L2 | NGSS MS-LS3-1, MS-LS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is a mutation, and what causes one?

2 Do you think most mutations are harmful, helpful, or have no effect? Why?

3 How might UV light from the sun affect your DNA?

4 Draw what you think happens to a protein when the gene coding for it has a mutation.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

UV Radiation Exposure | Cell Division Rate | Mutation Frequency | Protein Function Change | Organism Fitness

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Conditions

Moderate UV, moderate cell division

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High UV Exposure

UV Radiation at 90%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Rapid Division

Cell Division at 90%, moderate UV

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

DNA Repair Mechanisms — Cellular enzymes that detect and fix DNA damage before it becomes a permanent mutation

Chemical Mutagen Exposure — Chemicals in smoke, food, or the environment that damage DNA

Cancer Risk — The probability of uncontrolled cell growth caused by accumulated mutations in growth-control genes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens when DNA copies itself and makes a mistake? Use evidence from your simulation to support your answers.

1. **Why are most mutations neutral rather than harmful or beneficial?**

2. **How does the model explain why skin cancer risk increases with sun exposure?**

3. **If mutations are mostly harmful, why are they also considered the 'raw material for evolution'?**

4. **Why might rapidly dividing cells (like skin, gut lining, and blood cells) have higher cancer risk?**

5. **How does understanding mutation help explain why sunscreen is important, based on the model?**

8 STEM Challenge: Design a Skin Cancer Prevention Campaign

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YOUR ENGINEERING MISSION

Design an evidence-based public health campaign about UV exposure and skin cancer risk, using your model data to create clear, compelling messaging.

THE SCENARIO

A dermatology clinic has hired your team to create a skin cancer prevention campaign for teenagers. Your messaging must be based on scientific evidence from your mutation model, not scare tactics.

Engineering Design Thinking

1. WHAT KEY FACTS FROM YOUR MODEL WILL ANCHOR THE CAMPAIGN MESSAGING?
2. HOW WILL YOU COMMUNICATE RISK WITHOUT USING SCARE TACTICS?
3. WHAT SPECIFIC PROTECTIVE BEHAVIORS WILL YOUR CAMPAIGN RECOMMEND?
4. HOW WILL YOU MEASURE WHETHER THE CAMPAIGN CHANGES BEHAVIOR?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Genetic Counselors and Molecular Biologists study how mutations affect health and disease, helping families understand genetic risks and developing new therapies. They earn $65,000-$110,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Your model shows that when UV Radiation Exposure is set to 90%, Mutation Frequency increases dramatically and Organism Fitness declines. Which explanation is best supported by the model?

- **A.** UV radiation directly kills organisms by heating their bodies
- **B.** UV radiation damages DNA bases, leading to more copying errors during cell division, which produces more mutations that alter protein function and reduce fitness
- **C.** UV radiation makes organisms weaker by destroying their immune systems
- **D.** UV radiation causes organisms to stop reproducing entirely

2 The model shows that both UV Radiation Exposure and Cell Division Rate independently increase Mutation Frequency. A student asks why skin cells are at higher cancer risk than brain cells. Which model-based explanation is correct?

- **A.** Skin cells are larger than brain cells, making them easier targets for UV
- **B.** Skin cells are exposed to UV radiation AND divide rapidly, meaning both external inputs are high, compounding the mutation frequency
- **C.** Brain cells are immune to mutations because they are protected by the skull
- **D.** Skin cells have weaker DNA than brain cells

3 Your model shows that Protein Function Change has a POSITIVE/NEGATIVE (+/-) relationship with Organism Fitness. What does this dual relationship represent?

- **A.** Mutations always increase fitness in some organisms and decrease it in others
- **B.** Most mutations change protein function in ways that are harmful to the organism, but rare mutations can be beneficial, so the effect can go either direction
- **C.** Protein function changes are always neutral and do not affect fitness
- **D.** The +/- sign means the relationship is unknown and cannot be predicted

4 A student claims that UV radiation causes mutations instantly the moment it hits DNA. Using the model, which response is most accurate?

- **A.** The student is correct because UV immediately changes DNA into a mutated form
- **B.** UV radiation damages DNA bases immediately, but the damage becomes a mutation only when the cell divides and copies the damaged DNA incorrectly
- **C.** UV radiation has no effect on DNA and mutations are purely random
- **D.** UV radiation only affects RNA, not DNA

5 **Your model shows that mutations are the raw material for evolution, yet most mutations reduce organism fitness. How do these two facts coexist?**

A. They cannot coexist. If most mutations are harmful, evolution should not be possible.

B. Evolution works at the population level over many generations. Most mutations are harmful to individuals, but the rare beneficial mutation spreads through the population via natural selection, driving adaptation.

C. All mutations are beneficial for evolution because they create variety

D. Harmful mutations are actually beneficial in disguise because they force organisms to adapt